ARAVIND V BAYARI

Embedded Systems Veteran | 25+ Years Across Defence, HealthTech, IoT & Beyond

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PROFESSIONAL SUMMARY

Versatile embedded systems professional with 25+ years of hands-on experience across 10+ companies spanning Defence & Aerospace, HealthTech, Medical Devices, IoT/Wearables, Test & Measurement, Vehicle Telematics, Food Robotics, Industrial Automation, Semiconductor, and LED Lighting. Career arc from Engineer at System Controls—working on mission-critical test systems for HAL's LCA Tejas and GTRE's Kaveri Engine—through 8 years as Chief Engineer at Pyrodynamics designing the award-winning SCAD product line, to Project Lead at Axiom Consulting, Senior Engineer at Element14/Avid Technologies, and Architect at iTriangle Infotech. Co-founded CleaErr Tech Solutions / toN0tErr in 2017, providing end-to-end hardware consulting to startups in food robotics, semiconductors, and LED lighting. Currently serving as Hardware Lead at Kubo Care, designing radar-based AI health monitoring systems for senior living facilities. Recipient of the Technology Development Board Award for Successful Indigenization (SCAD 508, Rs. 5 Lakhs + Trophy, 2013).

PROFESSIONAL EXPERIENCE

Kubo Care Pvt Ltd
2024 – Present

Hardware Lead

Antler-backed HealthTech startup | $1M Seed (Jun 2024) | Bangalore & San Francisco

- Leading end-to-end hardware design for radar-based ambient AI health monitoring devices targeting senior living facilities and home care
- Architecting millimeter-wave radar systems (1–10 mW transmit power) for contactless vital signs monitoring, fall detection and prevention, sleep tracking, and environmental sensing
- Ensuring full regulatory compliance: FDA and FCC approval pathways for sensor subsystems; SOC2-compliant system architecture for data security
- Designed and delivered **Gen 1.0** hardware prototype; currently leading **Gen 2.0** hardware iteration with improved radar performance, miniaturization, and manufacturability
- Managing hardware team including embedded HW engineer; collaborating closely with ML, firmware, cloud, and software teams
- Addressing India's senior care market projected to grow from $12–15B to $40–50B by 2030

Tech: mmWave Radar, AI/ML Edge Processing, Embedded Linux, Sensor Fusion, FDA/FCC Compliance

CleaErr Tech Solutions / toN0tErr

Dec 2017 – Present

Co-founder & CTO

Embedded hardware consulting practice serving startups and SMEs

- Founded and operate an embedded systems consulting firm delivering concept-to-production hardware solutions for early-stage startups
- Managing full product lifecycle: requirements capture, architecture, schematic & PCB design, firmware, prototyping, testing, and production handoff

1. Freshot Robotics — Idli ATM (Feb 2019 – Oct 2020)

- Designed embedded electronics for an automated food dispensing machine (“Idli ATM”) — a first-of-its-kind food robotics product
- Developed motor control, sensor interfacing, HMI, and communication subsystems for autonomous food preparation and vending

Domain: Food Robotics & Automation

2. Tenxer Technologies (Dec 2017 – Feb 2019)

- Provided embedded hardware consulting for IC evaluation and semiconductor test platform development
- Designed evaluation boards and test fixtures for integrated circuit characterization

Domain: Semiconductor & IC Evaluation

3. SupraEnergy Lighting (May 2018)

- Designed LED driver circuits and lighting control electronics for commercial LED lighting products

Domain: LED Lighting

iTriangle Infotech Pvt Ltd

Aug 2016 – Nov 2017

Architect, Embedded Systems Design

Largest Indian manufacturer of Vehicle Telematics hardware

- Led embedded systems architecture for next-generation vehicle telematics products at India’s largest telematics hardware manufacturer
- Defined hardware architecture for GPS/GPRS/GSM-based vehicle tracking, fleet management, and driver behavior monitoring devices
- Worked on system-level design including power management, wireless communication stacks, and automotive-grade environmental considerations
- Collaborated with firmware and cloud teams to ensure seamless end-to-end telematics data pipeline from device to server

Tech: GPS, GPRS, GSM, CAN Bus, Automotive-grade Embedded Systems, Telematics Protocols

Element14 India / Avid Technologies

Aug 2015 – Jun 2016

Senior Embedded Systems Engineer

- Designed the **Thunderboard Wear** — an ARM mbed Wearable Reference Design Board showcasing Silicon Labs’ EFM32 ecosystem for IoT wearables
- Full hardware design: schematic capture in OrCAD Capture, multi-layer PCB layout in Cadence Allegro, BOM optimization, and design for manufacturing
- Integrated EFM32GG995 Giant Gecko MCU, memory LCD display, BLE BGM111 module, BMX055 9-axis IMU, Si1144 proximity/ambient light sensor
- Developed firmware on mbedOS; managed complete project lifecycle from concept through production and contract manufacturing
- Coordinated with Silicon Labs engineering teams for component selection, reference designs, and technical validation

Tech: EFM32GG995, memLCD, BLE BGM111, BMX055, Si1144, OrCAD Capture, Cadence Allegro, mbedOS, Simplicity Studio

Axiom Consulting Pvt Ltd

Aug 2011 – Jul 2015

Project Lead — Embedded Electronics

Design consultancy specializing in Medical Devices, IoT, and Industrial Systems

- Led 6 diverse embedded projects from concept through realization, spanning medical devices, IoT, and laboratory automation
- Managed cross-functional teams; drove certification processes (VDE/UL); handled vendor development and production coordination

1. Illumination Ring for Colposcopy Device

- Designed a high-intensity LED illumination ring for a cervical cancer screening colposcope
- 12x 1W high-power LEDs with precision PWM intensity control for uniform clinical illumination
- Interfaced PCA9685 16-channel PWM driver with 8051 microcontroller; designed constant-current LED drivers using NCP3066 and LXCL-EYW4 LEDs

Tech: PCA9685, 8051, NCP3066, LXCL-EYW4, PWM LED Drivers

2. CPAP — Continuous Positive Airway Pressure Device

- Designed embedded electronics for a sleep apnea ventilation device (CPAP machine)
- Implemented precision BLDC motor control using DRV8312 three-phase driver for blower regulation
- Integrated MPX2010 pressure sensor for real-time airway pressure monitoring and closed-loop control
- MSP430F5529 ultra-low-power MCU for system control, patient data logging, and compliance monitoring

Tech: DRV8312, MPX2010, MSP430F5529, BLDC Motor Control, Pressure Sensing

3. Stempeutron — Automated SVF Cell Isolation System

- Designed automation electronics for a medical device that isolates Stromal Vascular Fraction (SVF) cells from adipose tissue
- Developed for Stempeutics Research (Manipal Education / Cipla Group) for point-of-care regenerative medicine
- Implemented precision motor control, temperature regulation, fluid handling, and process sequencing for automated cell isolation

Domain: Medical Devices, Regenerative Medicine, Lab Automation

4. Package Shock Logger

- Designed a ruggedized GPS/GPRS-enabled shock and vibration data logger for transit package monitoring
- IP65-rated enclosure for harsh logistics environments; battery-optimized for extended field deployment
- Integrated ADXL345 3-axis MEMS accelerometer for multi-threshold shock event capture and logging
- SIM908 GPS/GPRS combo module for real-time location tracking and remote data upload; ATmega328 MCU for data processing

Tech: ADXL345, SIM908, ATmega328, GPS, GPRS, IP65, Low-Power Design

5. Lab Digitization

- Developed a comprehensive LabVIEW-based framework for laboratory instrument digitization and data management
- Automated data acquisition, instrument control, and report generation for analytical and research laboratories

Tech: LabVIEW, Data Acquisition, Instrument Control, GPIB/RS-232

6. Smart Shelf — UWB Real-Time Location System

- Developed an Ultra-Wideband (UWB) based real-time location system (RTLS) for chemical laboratory inventory management
- Leveraged DecaWave UWB modules for centimeter-level precision asset tracking on laboratory shelving
- Enabled real-time tracking of hazardous chemical containers for safety compliance and audit readiness

Tech: DecaWave DW1000, UWB RTLS, Precision Indoor Positioning

SoftApps

Aug 2010 – Aug 2011

Senior Engineer / Tech Lead

- Designed a microcontroller-based Programmable Logic Controller (PLC) for cement plant process automation
- Developed Modbus-based GSM module for industrial SMS alerting, enabling remote monitoring and alarm notification for plant operators
- Implemented Modbus RTU/TCP communication protocols for integration with SCADA systems and existing plant infrastructure

Tech: Microcontroller PLC, Modbus RTU/TCP, GSM, SCADA Integration, Industrial Automation

Pyrodynamics

Jul 2002 – Jul 2010

Chief Engineer (8 Years)

Only Indian company designing world-class strain measurement systems — competing with global manufacturers

- Served as the primary hardware and electronics engineer for 8 years, designing the entire SCAD product line from ground up
- Responsible for analog signal conditioning, data acquisition, digital control, embedded firmware, PCB design, and production engineering
- Products deployed across India's premier national laboratories, research institutions, and heavy engineering organizations

SCAD 500 — Strain Measurement System

- Flagship multi-channel strain measurement and data acquisition system
- Deployed at BHEL (Bharat Heavy Electricals Ltd) and BARC (Bhabha Atomic Research Centre)
- Exhibited in Germany — only Indian system showcased alongside global strain measurement manufacturers

SCAD 500L — Laboratory Variant

- Laboratory-optimized variant of SCAD 500 with enhanced precision and noise performance
- Deployed at NML Jamshedpur (National Metallurgical Laboratory) for materials research

SCAD 508 — Award-Winning Indigenized System

- Advanced 8-channel strain measurement system with indigenous design and manufacturing
- Deployed at Jadavpur University for structural engineering research
- **Won the Technology Development Board Award for Successful Indigenization (2013)** — Rs. 5 Lakhs cash prize + Trophy from Dept. of Science & Technology, Government of India

SCAD 100, SCAD 200 & SCAD Cal

- SCAD 100: Compact single/dual-channel strain indicator for field measurements
- SCAD 200: Mid-range multi-channel system for laboratory and field applications
- SCAD Cal: Precision calibration unit for the entire SCAD product family

Phase Shifter for ISI SYS Germany

- Designed a precision electronic phase shifter module for shearography (optical non-destructive testing) systems
- Delivered to ISI SYS Germany — international export of Indian-designed precision instrumentation

Tech: Analog Signal Conditioning, Strain Gauge Bridge Circuits, Multi-channel DAQ, Precision ADC/DAC, Embedded Firmware, PCB Design

System Controls

Aug 2000 – Jun 2002

Engineer — Defence & Aerospace Test Systems

Specialized in test benches and data acquisition systems for India's Defence and Aerospace organizations

- Designed and delivered 7 embedded test and measurement systems for premier Indian defence and aerospace establishments

1. Control Stick Test Bench — HAL, LCA Tejas

- Designed and built a test bench for evaluating the flight control stick assembly of HAL's Light Combat Aircraft (LCA) Tejas
- Precision force/displacement measurement, data acquisition, and automated test sequencing

2. Tachogenerator Test Bench — Southern Electronics

- Built an automated test bench for characterizing tachogenerators used in defence servo systems
- Speed-voltage linearity testing, ripple analysis, and temperature coefficient measurement

3. Portable Data Acquisition System — NAL

- Designed a portable multi-channel data acquisition system for the National Aerospace Laboratories (NAL)
- Field-deployable unit for structural testing and flight test instrumentation

4. Multi-channel Filter — ADA

- Designed a programmable multi-channel analog filter system for the Aeronautical Development Agency (ADA)
- Signal conditioning for flight test data acquisition and telemetry systems

5. Servo Valve Simulation — GTRE, Kaveri Engine

- Developed a servo valve simulation system for GTRE's Kaveri turbofan engine development program
- Real-time simulation of hydraulic servo valve dynamics for engine fuel control system testing

6. AOA Vane Calibrator

- Designed precision calibration equipment for Angle of Attack (AOA) vanes used in flight test instrumentation
- Angular positioning accuracy and repeatability verification for air data measurement systems

7. DC Servo & Stepper Motor Drives

- Designed DC servo motor drive electronics and stepper motor controllers for test automation and positioning systems
- Closed-loop position/velocity control with encoder feedback for precision mechanical test setups

Tech: Data Acquisition, Analog Signal Conditioning, Motor Drives (DC Servo, Stepper), LabVIEW, Instrumentation

DOMAIN EXPERTISE MATRIX

Domain	Experience	Key Companies / Projects
HealthTech	2024–Present	Kubo Care — Radar-based AI health monitoring for senior care
Medical Devices	2011–2015	Axiom — CPAP, Stempeutron (SVF cell isolation), Colposcopy illumination
Defence & Aerospace	2000–2002	System Controls — HAL LCA Tejas, GTRE Kaveri, NAL, ADA
Test & Measurement	2002–2010	Pyrodynamics — SCAD product line (500, 500L, 508, 100, 200, Cal)
IoT / Wearables	2011–2016	Element14 — Thunderboard Wear; Axiom — Package Shock Logger, Smart Shelf
Vehicle Telematics	2016–2017	iTriangle Infotech — GPS/GPRS telematics hardware architecture
Food Robotics	2019–2020	CleaErr / Freshot Robotics — Idli ATM automated food dispensing
Industrial Automation	2010–2011	SoftApps — Microcontroller PLC for cement plants, Modbus GSM alerts
Semiconductor	2017–2019	CleaErr / Tenxer Technologies — IC evaluation platforms
LED / Lighting	2018	CleaErr / SupraEnergy Lighting — LED driver design
Smart Storage / RTLS	2011–2015	Axiom — UWB Smart Shelf (DecaWave) for chemical lab tracking

KEY SKILLS

Technology: Analog/Digital/Mixed Signal Design, PCB Layout (2-layer to 8+ layer), Signal Integrity, Power Electronics, EMI/EMC (IEC/EN/ANSI), Radar (mmWave), Strain Gauge Instrumentation, Precision Data Acquisition

Mechatronics: DC Servo Drives, Stepper Motor Control, BLDC Motor Control (DRV8312), Sensor Fusion, Closed-loop Control Systems, Pneumatics/Hydraulics Interface

Architecture: System-level Hardware Architecture, SoC/SoM Selection, RF & Wireless (BLE, WiFi, GPS, GPRS, UWB, mmWave), Multi-board System Design, DFM/DFT

Firmware: mbedOS, Keil ARM, Simplicity Studio, GCC ARM, Arduino, CooCox, IAR, MSP430, 8051, AVR, PIC, Embedded Linux

Testing & Certification: FDA, FCC, VDE, UL, SOC2, IEC 60601 (Medical), IP65/IP67 Environmental, Automotive-grade Qualification

Tools: OrCAD Capture, Cadence Allegro, LabVIEW, Simplicity Studio, MATLAB, Oscilloscopes, Logic Analyzers, Spectrum Analyzers, Network Analyzers

Management: Product Lifecycle Management, Project Leadership, Vendor Development, Contract Manufacturing, Cross-functional Team Leadership, Startup Co-founding, Client Engagement & Delivery

EDUCATION

Bachelor of Engineering — Instrumentation & Electronics Engineering
RV College of Engineering, Bangalore University (1995–1999)

AWARDS & RECOGNITION

Technology Development Board Award for Successful Indigenization (2013) — Rs. 5 Lakhs cash prize + Trophy, awarded by the Technology Development Board, Department of Science & Technology, Government of India, for the SCAD 508 Strain Measurement System. Recognized for successfully indigenizing world-class test & measurement instrumentation.

International Exhibition — SCAD 500 in Germany — The SCAD 500 strain measurement system was exhibited at an international trade show in Germany, representing India as the only indigenous system showcased alongside established global strain measurement manufacturers.

NOTABLE CLIENTS & INSTITUTIONS SERVED

Defence & Aerospace: HAL (Hindustan Aeronautics — LCA Tejas), GTRE (Gas Turbine Research Establishment — Kaveri Engine), NAL (National Aerospace Laboratories), ADA (Aeronautical Development Agency), Southern Electronics

National Laboratories & Research: BARC (Bhabha Atomic Research Centre), BHEL (Bharat Heavy Electricals Limited), NML Jamshedpur (National Metallurgical Laboratory), Jadavpur University

International: ISI SYS Germany (Shearography / Optical NDT Systems)

Healthcare & Biotech: Stempeutics Research (Manipal Education / Cipla Group), Kubo Care (Antler-backed HealthTech)

Startups & SMEs: Freshot Robotics, Tenxer Technologies, SupraEnergy Lighting, iTriangle Infotech